Prescriptive ground reaction force characterization of a transtibial prosthesis gait simulator driven by stature-based trajectories

Luis M. Plasencia[1], Leonardo F. Machiavello[1], Adrian V. Gálvez[1], Victoria E. Abarca[1*], Dante A. Elias[1]

[1*]Biomechanics and Applied Robotics Research Laboratory, Pontificia Universidad Católica del Perú, 1801 Universitaria Avenue, San Miguel, 15088, Lima, Peru.

*Corresponding author(s). E-mail(s): victoria.abarca@pucp.edu.pe;
Contributing authors: luis.plasencia@pucp.edu.pe; leonardo.machiavello@pucp.edu.pe; adrian.vilchez@pucp.edu.pe; delias@pucp.pe;

Abstract

Laboratory evaluation of transtibial prostheses requires reproducible gait kinematics and ground reaction forces (GRF) without recruiting amputees at every session. Motion-constrained gait simulators execute a prescribed trajectory, typically from motion capture, but do not generate one. How GRF depends on a simulator's controllable inputs is rarely characterized. A stature-driven generator built on published models supplies a complete stance-phase trajectory without amputee training data. A three-degree-of-freedom simulator executes trajectories on a force platform. A GRF protocol relates vertical offset to impulse-equivalent mass across statures, enabling prescriptive loading with position control. Leave-one-subject-out cross-validation on 47 Kuopio subjects gave a normalized root-mean-square error of 0.97 to 1.39 across five trajectory variables. Same-fold population-mean baselines, unscaled or scaled by stature, had lower error, 0.87 to 0.93. The generator's role is to supply an executable trajectory for each target stature. External validation on 22 Camargo subjects gave 1.82 to 2.86 on these variables. An additive model fitted to 45 stature-offset combinations yielded slopes of $\mathbf{-0.50}$ kg per centimeter of stature and $\mathbf{+3.27}$ kg per millimeter of offset, with $\boldsymbol{R^2 = 0.993}$ and residual standard error 0.49 kg. Median intra-session coefficient of variation was 0.17 %. Inter-session intraclass correlation coefficient was 0.996. Limits of agreement were $\mathbf{-0.988}$ to $\mathbf{+1.148}$ kg. A

60 kg prescriptive session showed a **+2.1** kg mean bias across eight calibration-grid statures (164–180 cm). A between-session tare shift explained approximately 1.2 kg. Median intra-condition standard deviation was 0.06 kg. Researchers can prescribe a repeatable, stature-specific impulse-equivalent load for prosthetic evaluation.

Keywords: gait simulator, ground reaction force, prosthesis evaluation, trajectory generation, transtibial prosthesis

1 Introduction

Lower-limb amputation restricts mobility and quality of life. An estimated 2.3 million individuals were living with limb loss in the United States as of 2019 (Rivera, Churovich, Anderson, & Potter, 2024). Prosthetic limbs restore ambulatory function for many of these individuals, although with a higher metabolic cost than in non-amputees (Waters & Mulroy, 1999).

Prosthesis simulators worn by non-disabled participants offer an alternative to testing with individuals with limb loss (Neelen, van der Windt, Major, & Smit, 2026). Most of these simulators are transfemoral, and the effect of transtibial simulators on gait has been little evaluated (Neelen et al., 2026). Stationary mechanical rigs mount and move an actual prosthetic device without a human wearer. A six-degree-of-freedom parallel robot with iterative learning control, for example, actively tracks a target ground reaction force (GRF) using motion-capture and force-plate data from a single transtibial amputee as reference. It reduces the root-mean-square error between simulated and real vertical GRF to 35 N after six control iterations (Aubin, Cowley, & Ledoux, 2008). Stationary robotic test rigs also load a prosthetic joint under controlled conditions (Etoundi et al., 2022). A three-degree-of-freedom gait simulator driven by predefined hip trajectories has been used to evaluate passive and microprocessor-controlled prosthetic knees without human participants (N.J. Kim, Lee, Oh, & Kim, 2026).

The dynamic loading profile of a transtibial prosthesis consists of the resultant vertical and horizontal GRF vector and the shank inclination angle from heel contact to toe-off. ISO 22675:2024 specifies standardized cyclic testing procedures for ankle-foot devices to evaluate their structural strength and durability (*Prosthetics – Testing of ankle-foot devices and foot units – Requirements and test methods*, 2024). Device-level loading is also linked to long-term outcomes, including contralateral knee osteoarthritis and residual-limb tissue injury at the socket interface, so it should be characterized during design and testing (Graser, Day, & Buis, 2020; Knight, Dearth, & Hendershot, 2021; Slater, Halsne, Czerniecki, & Morgenroth, 2024).

The gait simulator used in this work is motion-constrained, not force-controlled, so it cannot directly prescribe GRF. The relationship between its motion parameters and the resulting GRF must therefore be determined experimentally. N.J. Kim et al. (2026) adjusted the vertical actuator displacement of their simulator until a target load was reached against a rigid surface, at two load levels. To our knowledge, no published

protocol models the GRF generated by a motion-constrained prosthetic gait simulator during stance as a function of a controllable input across a range of loads. The closest related work varies transtibial prosthetic alignment, including a vertical displacement of up to 20 mm, and measures GRF and whole-body angular momentum during gait and sit-to-stand transitions (Nolasco, Silverman, & Gates, 2023). Its participants were amputees, and no simulator was used. Peak vertical GRF was not significantly affected by alignment as a main effect, and only inter-limb loading asymmetry shifted between conditions (Nolasco et al., 2023). Because the participants could compensate through gait adaptation, that result does not transfer to a motion-constrained simulator.

A motion-constrained simulator also requires a prescribed trajectory, which, together with the tested component, determines its mechanical output. Approaches to generating representative gait trajectories for prosthetic device evaluation include data-driven models trained on motion-capture recordings from real participants (Bicer, Phillips, Melis, McGregor, & Modenese, 2024; M. Kim & Hargrove, 2023; Tan et al., 2026). They also include closed-form analytical models that derive joint kinematics from walking speed and anthropometric parameters (Koopman, van Asseldonk, & van der Kooij, 2014). The latter approach suits prosthetic testing because it avoids the recruitment constraints that have limited most published prosthesis-testing protocols to single subjects or small cohorts (Kooiman et al., 2023). Because the trajectory is generated and not taken from recorded participants, a test protocol is not limited to the statures of a single reference dataset. Data-driven trajectory generators, in contrast, need training data large enough to represent the target population. Individuals with lower-limb loss are underrepresented in public motion-capture repositories, so adapting these models to prosthetic use has required purpose-built generative augmentation (M. Kim & Hargrove, 2023).

This work aims to characterize the ground reaction force (GRF) resulting from the interaction between a simulator-driven prosthesis and the supporting surface during stance. To supply its trajectories without new participant recordings, we adopt a stature-driven gait trajectory generator based on the closed-form analytical model of Koopman et al. (2014). The generator takes subject stature as its sole input and produces a complete lower-limb kinematic trajectory, calibrated and corrected using leave-one-subject-out cross-validation on a motion-capture dataset of $N = 47$ subjects. Stature acts through segment lengths, cycle timing, and the hip trajectory, not through the joint-angle profile. In that cross-validation, the generator was no more accurate than the population mean. The main contribution is an empirical protocol that characterizes the GRF resulting from prosthetic foot–surface interaction in a motion-constrained simulator as a function of vertical offset. The resulting impulse-based model can be inverted to determine the offset required to achieve a target impulse-equivalent mass for a given stature. An independent prescriptive session measures the accuracy of that prescription against the target mass and its repeatability across trials. A control-layer integration allows a single physical simulator to run stature- and offset-specific trajectories without manual reconfiguration.

The generator and the protocol form a workflow for configuring a motion-constrained prosthetic gait simulator to a target stature and impulse-equivalent mass. It gives research groups without force-controlled infrastructure a calibrated way to

reproduce stature-specific vertical loading, and a starting point for closed-loop force control.

2 Methodology

2.1 Gait trajectory generator

A five-step pipeline generates a simulator trajectory based on participant stature. Step 1 estimates segment lengths from stature. Step 2 estimates walking speed and cycle duration. Step 3 predicts joint angles using a model selected from twelve published candidates, as described in Section 2.5.1, and corrects their amplitudes using an affine calibration fitted by leave-one-subject-out cross-validation (LOSO). Step 4 computes the knee and ankle positions using double-pendulum kinematics and corrects them. Step 5 exports the stance phase as a position-command file. Stature is the sole input to the kinematic trajectory. However, the exported trajectory also depends on the mounting-point distance d, defined in Section 2.7, which is a physical setting independent of stature.

2.2 Trajectory accuracy metrics

Agreement between predicted and reference kinematics is quantified by Pearson's r, which measures waveform shape independently of amplitude and offset, and by the root-mean-square error (RMSE) in the original units of each coordinate. Because RMSE values in centimeters or degrees are not directly comparable across joints and datasets with different inter-subject variability, a normalized variant is also reported, as defined in Eq. (1),

$$\mathrm{RMSEnorm} = \sqrt{\frac{1}{M}\sum_{t=1}^{M}\left(\frac{\hat{x}(t) - x(t)}{\sigma(t)}\right)^2} \tag{1}$$

where $\hat{x}(t)$ and $x(t)$ are the predicted and reference coordinates at gait-cycle instant t, $\sigma(t)$ is the between-subject standard deviation of the reference at that instant, and $M = 101$ is the number of normalized gait-cycle points (0%–100%). An RMSEnorm of 1.0 indicates that the root-mean-square prediction error, normalized by the between-subject standard deviation at each gait-cycle instant, equals 1.0. The thresholds follow the criterion of Moissenet, Leboeuf, and Armand (2019), who compared prediction error against a corridor of one between-subject standard deviation. RMSEnorm at most 1.0 is Excellent, 1.0–1.5 Good, 1.5–2.0 Acceptable, and above 2.0 Deficient. Because the denominator is the between-subject standard deviation of the reference population, a leave-one-subject-out population-mean predictor reaches RMSEnorm close to 1.0 for the Kuopio dataset. This baseline is therefore reported alongside the generator. A second baseline scales that mean by the stature of the left-out subject and keeps its shank length constant over the cycle.

2.3 Segmental anthropometry

Thigh and shank segment lengths are estimated from stature H using the proportionality constants reported by Winter (2009), $L_{\text{thigh}} = 0.245\,H$ and $L_{\text{shank}} = 0.246\,H$.

2.4 Walking speed and cycle duration

Self-selected walking speed is estimated from the Froude number using Eq. (2) (Leurs et al., 2011),

$$v = \sqrt{Fr \cdot g \cdot L}, \qquad L = 0.530\,H, \qquad Fr = 0.25 \tag{2}$$

where $Fr = 0.25$ is the dimensionless speed at which walking is energetically optimal, L is the leg-length fraction of stature from Winter (2009), and g is gravitational acceleration. Speeds are reported in km/h. Cycle duration is obtained by evaluating Equation 3 of Koopman et al. (2014), a step-ratio linear regression on speed and stature, at the Froude-estimated speed and subject stature. It was used for all three candidate angle models because it is the only published regression that relates cycle duration to both speed and stature. That regression was validated for speeds of 0.5–5.0 km/h. Under Eq. (2), the estimated speed exceeds 5.0 km/h for statures above approximately 148 cm, so the regression is extrapolated for taller subjects.

2.5 Joint angle prediction

2.5.1 Candidate model selection

Two inclusion criteria were fixed before evaluating results. The first required previously fitted and published regression coefficients that could be reused without retraining. The second required inputs limited to static anthropometry and walking speed or cadence, both of which the pipeline estimates from stature. Table 1 lists the twelve candidates and identifies the criterion not met by each excluded model.

Of the four remaining candidates, one (Romero-Sorozabal et al., 2024) was excluded because its published stature coefficient for knee and ankle vertical position yields a segment length approximately twice that expected from standard anthropometric proportions. The three retained candidates, Koopman et al. (2014), Zhao et al. (2026), and Yun et al. (2014), were evaluated against three independent reference datasets, each covering a different set of outputs. Senden, Marcellis, Willems, Witlox, and Meijer (2024) provide a mean knee flexion curve for the 21 men aged 18 to 29 years who walked at a comfortable speed on a treadmill. The candidates were compared against this group-mean curve at the group's mean stature of 182 cm. Body mass was set to the group's mean of 81.0 kg. The Ferber, Brett, Fukuchi, Hettinga, and Osis (2024) treadmill dataset provided knee position for 40 participants, selected as described in Section 2.10.1. Lavikainen et al. (2024) provide ankle position and shank angle for 47 participants walking overground. Each candidate's published equations were applied to the reported statures and speeds. For comparisons of positions in Tables 2 and 3, joint angles were converted to Cartesian coordinates using double-pendulum forward kinematics.

Table 1: Twelve published candidate models for joint-angle or position prediction from anthropometry, screened against the two inclusion criteria above.

#	Candidate	Reason excluded
1	Koopman et al. (2014)	Met both criteria
2	Zhao et al. (2026)	Met both criteria
3	Yun et al. (2014)	Met both criteria
4	Romero-Sorozabal, Delgado-Oleas, Laudanski, Gutiérrez, and Rocon (2024)	Met both criteria
5	Semwal, Jain, Maheshwari, and Khatwani (2023)	LSTM (long short-term memory) + CNN (convolutional neural network); trained weights not published
6	Xin et al. (2025)	Gaussian-process regression trained on 40 participants; training data and fitted model not published
7	Hu, Shen, Zhao, Qu, and Ye (2020)	LASSO (least absolute shrinkage and selection operator) regression coefficients not published
8	Luu, Low, Qu, Lim, and Hoon (2014)	Requires the full training dataset (600 trials, 70 subjects); not reusable from published parameters alone
9	Wu, Liu, Liu, Chen, and Guo (2018)	Autoencoder and Gaussian-process weight matrices not published
10	Luu, Lim, Qu, Hoon, and Low (2011)	Trained neural-network weights not published
11	Shkedy Rabani, Mizrachi, Sawicki, and Riemer (2022)	Personalizes by terrain slope, not by anthropometry
12	Al Kouzbary et al. (2026)	Requires the measured shank angle as input, not static anthropometry

Koopman et al. (2014) model four joint angles, hip ab-/adduction, hip flexion/extension, knee flexion/extension, and ankle plantar-/dorsiflexion, as piecewise quintic Hermite splines through six gait-cycle events, with spline knot positions and magnitudes predicted by published linear regressions on walking speed and stature. In the generator, these regressions are evaluated at a fixed reference speed of 5.0 km/h, the upper end of their validated speed range. Their stature input is fixed at a reference adult stature of 173.5 cm. With subject-specific inputs, the regressions predicted a dependence of the shank angle on stature that the reference data did not show. Stature therefore enters the generator through segment lengths, cycle duration, and the hip trajectory, not through the angle profile.

The shank inclination angle θ_{shank}, shown in Fig. 1, is the sagittal-plane angle of the knee-to-ankle segment measured from the vertical through the knee. It is positive when the ankle lies ahead of the knee in the direction of walking and negative when it lies behind. On the simulator, the same value sets the tilt of the shank inclination axis, whose plate is perpendicular to the shank. A plate tilt of θ_{shank} from the horizontal therefore matches a shank angle of θ_{shank} from the vertical, with the same sign. The shank angle executed by the simulator is not published directly by Koopman et al.; instead, it is derived geometrically assuming a vertical pelvis, i.e., $\theta_{\text{thigh}} \approx \varphi_{\text{hip}}$ (Zhao et al., 2026), as

$$\theta_{\text{shank}} = \theta_{\text{thigh}} - \varphi_{\text{knee}} \tag{3}$$

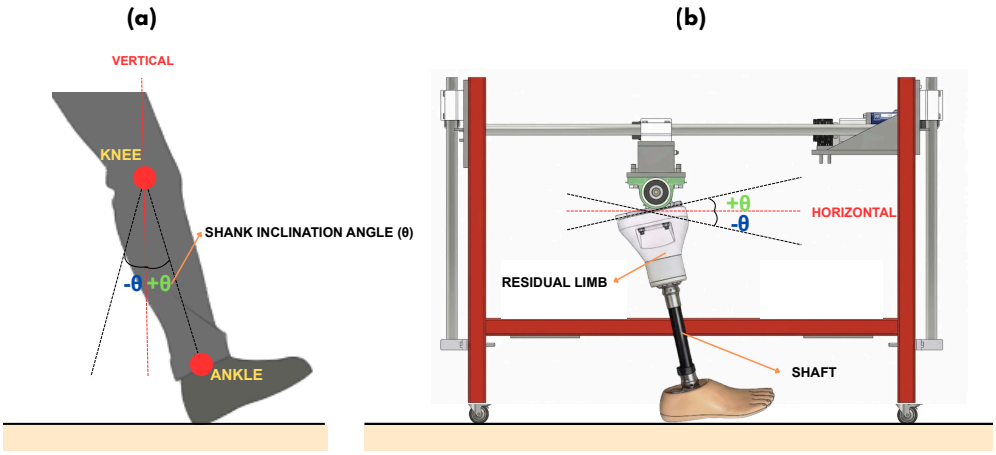


Fig. 1: Shank inclination angle θ_{shank}. (a) In the person, measured from the vertical through the knee. (b) On the simulator, measured from the horizontal

The geometric relationship in Eq. (3) was evaluated against an alternative approach based on the ankle angle, using the measured shank angles from the 47 Kuopio subjects. The alternative approach was rejected. Without calibration, the knee route gave $r = 0.990 \pm 0.009$ over the full cycle, whereas the ankle route, which assumes a flat foot throughout the cycle, gave $r = -0.35 \pm 0.04$ and inverted the sign during swing in all 47 subjects, with $r = -0.84 \pm 0.02$ in that phase.

Zhao et al. (2026) and Yun et al. (2014) carry a 50%-of-gait-cycle phase offset in their laterality parameter, so their performance changed with each dataset's side convention. Their values in Table 2 use the favorable convention, chosen after the fact for each dataset. Under the native convention of Table 3, both candidates underperform Koopman et al., including a sign inversion on shank angle. The model of Koopman et al. was selected because it does not depend on an after-the-fact convention choice.

Table 2: Correlation (r) of the three candidate models against measured knee flexion/extension angle and position, by reference source. The Ferber et al. (2024) dataset was recorded on a treadmill; r_x and r_y denote the horizontal and vertical components of knee position, respectively. [a] Zhao and Yun values reflect the favorable side of the laterality phase offset.

Source	n	Koopman	Zhao	Yun
Senden et al. (2024)	21	0.913	0.982[a]	0.939[a]
Ferber et al. (2024) (r_x)	40	0.946	0.947[a]	0.902
Ferber et al. (2024) (r_y)	40	0.793	0.815[a]	−0.698

Table 3: Correlation (r) of the three candidate models against ankle position (r_x: horizontal; r_y: vertical) and shank angle, Lavikainen et al. (2024) dataset ($n = 47$, overground). Native laterality configuration (no phase-offset adjustment applied).

Segment	Koopman	Zhao	Yun
Ankle r_x	0.999	0.975	0.990
Ankle r_y	0.979	0.918	0.837
Shank angle	0.990	−0.297	−0.277

2.5.2 Angle calibration

The Koopman model was derived from $N = 15$ participants with a mean stature of 1.69 ± 0.10 m. When applied without adjustment to the Kuopio Gait Dataset (Lavikainen et al., 2024) ($n = 47$, overground), it systematically overestimated angular amplitude. The thigh angle and the shank angle derived via Eq. (3) exhibit separate amplitude biases. Therefore, an affine correction was fitted independently to each angle using LOSO cross-validation over the same dataset. The affine transformation is defined by Eq. (4), where all quantities are in degrees.

$$\hat{\theta} = a\,\theta_{\text{Koopman}} + b \tag{4}$$

The coefficients (a, b) of Eq. (4) were estimated by minimizing root-mean-square error against the corresponding reference angle from the remaining $n - 1$ subjects. Table 4 reports the mean of the 47 leave-one-out pairs; a separate fit on the complete $n = 47$ dataset agrees with this mean to four decimal places and provides the coefficients applied to new subjects. The coefficients in Table 4 were derived from participants with statures of 161–187 cm (Lavikainen et al., 2024), so statures outside this range extrapolate the calibration.

Table 4: Mean leave-one-out affine calibration coefficients for the two corrected angles ($n = 47$, Lavikainen et al. 2024).

Segment	Gain (a)	Intercept (b, °)
Thigh	0.694	−1.59
Shank angle*	0.774	−11.58

* Derived via Eq. (3) (thigh − knee); not a direct Koopman output.

2.6 Forward kinematics and position correction

The kinematic chain starts at the hip. Within each gait cycle of duration T_{cycle} from Section 2.4, its horizontal coordinate, Eq. (5), advances at the mean walking speed v

given by Eq. (2).

$$x_h(t) = v\,t, \qquad t \in [0,\, T_{\text{cycle}}] \tag{5}$$

Its vertical coordinate, Eq. (6), follows the double-hump pelvis oscillation of normal gait (Luciano, Ruggiero, Minetti, & Pavei, 2022; Saunders, Inman, & Eberhart, 1953), where the vertical origin is placed at the hip ($y_0 = 0$), so all downstream joint positions are expressed relative to hip height.

$$y_h(t) = -A\cos\left(\frac{4\pi\,t}{T_{\text{cycle}}}\right), \qquad A \approx 2.25 \text{ cm} \tag{6}$$

The half-amplitude A is set to half the peak-to-peak vertical excursion of the center of mass at a freely selected walking speed, reported as 4–5 cm by Saunders et al. (1953). The midpoint of that range gives $A = 4.5/2 = 2.25$ cm.

The calibrated angles from Section 2.5.2 are used in the double-pendulum forward kinematics described by Eqs. (7)–(8) to compute the knee and ankle coordinates. Here, θ_{thigh} and θ_{shank} denote the calibrated angles, with θ_{shank} derived via Eq. (3); L_{thigh} and L_{shank} denote the segment lengths from Section 2.3. Positive θ denotes anterior tilt, and positive y points upward. Because the same kinematic equations apply throughout the gait cycle without switching between stance and swing, no phase-switching discontinuities are introduced into the reference trajectory.

$$x_k = x_h + L_{\text{thigh}}\sin\theta_{\text{thigh}}, \qquad y_k = y_h - L_{\text{thigh}}\cos\theta_{\text{thigh}} \tag{7}$$

$$x_a = x_k + L_{\text{shank}}\sin\theta_{\text{shank}}, \qquad y_a = y_k - L_{\text{shank}}\cos\theta_{\text{shank}} \tag{8}$$

Even after angle calibration, the raw double-pendulum positions exhibit two distinct problems. In the horizontal coordinate of Eqs. (7)–(8), when a segment's angular velocity is large and directed opposite to the advance direction, the rotational contribution $L_{\text{thigh}}\cos\theta_{\text{thigh}}\,\dot{\theta}_{\text{thigh}}$ briefly exceeds v, causing the horizontal position to move backward in short steps that cannot occur in overground walking. Such backward steps were observed for some subjects even within the validated stature range. In the vertical direction, a point-wise amplitude mismatch persists that cannot be resolved by reordering samples in time. The hybrid correction addresses these two problems separately.

Horizontal position is corrected by temporal warping (Ramsay & Silverman, 2005). The Fourier amplitude correction used for the vertical axis was initially tested for the horizontal coordinate but discarded because it introduced backward steps in trajectories that previously had none: 0 to 4 per hundred for knee X and 11 to 22 per hundred for ankle X. Warping evaluates the raw trajectory at a warped time $\tau(t)$, changing the timing of its values without altering their magnitudes. A monotone raw trajectory therefore stays monotone as long as the warp rate $\dot{\tau}$ is positive. The rate is parametrized as $e^{\gamma(t)}$, which keeps it positive without constraints on the optimizer and gives Eq. (9),

$$\frac{d\tau}{dt} = e^{\gamma(t)}, \quad \tau(0) = 0, \quad \tau(100) = 100 \tag{9}$$

where γ is a Fourier series of $K_w = 2$ harmonics, the simplest level at which LOSO error plateaus in a sweep from $K_w = 1$ to $K_w = 12$, and $\tau(100) = 100$ is enforced by

rescaling the cumulative integral. The corrected coordinate then takes the form

$$x_{\mathrm{corr}}(t) = c_0 + c_1\, x_{\mathrm{raw}}(\tau(t)) \tag{10}$$

In Eq. (10), c_0 and c_1 are constants fitted by leave-one-out regression. Adding a speed-derived covariate reduced RMSE but disrupted the monotonic relationship between stature and the output, as cycle-end displacement showed no correlation with stature in the Kuopio dataset ($r = -0.01$). The affine map is therefore kept constant by design. As a last step applied to every output, the Pool Adjacent Violators Algorithm (Ayer, Brunk, Ewing, Reid, & Silverman, 1955; Barlow, Bartholomew, Bremner, & Brunk, 1972), hereafter PAVA, projects the output onto the nearest non-decreasing sequence in the least-squares sense. Three of 47 Kuopio subjects at statures 161–165 cm retained backward steps in ankle X after warping because their raw pendulum output already contained them. PAVA resolves these without affecting already-monotone trajectories.

The vertical coordinate has no monotonicity constraint, so it needs only a point-wise amplitude adjustment. The corrected vertical position takes the form of Eq. (11),

$$y_{\mathrm{corr}}(t) = \alpha(t) + \beta(t)\, y_{\mathrm{raw}}(t) \tag{11}$$

where $\alpha(t)$ and $\beta(t)$ are independent Fourier series of K harmonics, with all coefficients solved jointly by ordinary least squares in the LOSO scheme. In a LOSO sweep over the Kuopio data, the cross-validation error reached a plateau at $K \approx 12$–14. The order $K = 14$ was adopted.

Because the four corrected coordinates are fitted independently, the shank length is no longer constant after correction and varies by up to 8 % across the cycle under default application settings. A midpoint projection restores it. The midpoint $\vec{m}$ and unit direction $\vec{u}$ of the corrected segment are given by Eq. (12), and both endpoints are reset symmetrically about $\vec{m}$ by Eq. (13).

$$\vec{m} = \tfrac{1}{2}[(x_k, y_k) + (x_a, y_a)]\,, \quad \vec{u} = \frac{(x_k, y_k) - (x_a, y_a)}{\|(x_k, y_k) - (x_a, y_a)\|} \tag{12}$$

$$(x_k, y_k)' = \vec{m} + \tfrac{L_{\mathrm{shank}}}{2}\,\vec{u}, \quad (x_a, y_a)' = \vec{m} - \tfrac{L_{\mathrm{shank}}}{2}\,\vec{u} \tag{13}$$

Both endpoints move by the same amount because there is no physical reason to fix one of them. The projection leaves the uncorrected double-pendulum output unchanged, because that output is already rigid. For the Kuopio dataset, this correction slightly reduces vertical accuracy, as shown in Section 3.1.

2.7 Prosthetic mounting-point projection

The simulator follows the prosthetic coupling point and not the anatomical ankle. This point lies on the corrected shank segment at a distance d from the ankle toward the knee. For a given d, the mounting-point position is defined by Eq. (14), where $\vec{P}_a$

and $\vec{P}_k$ denote the corrected ankle and knee positions.

$$\vec{P}_m = \vec{P}_a + d\,\frac{\vec{P}_k - \vec{P}_a}{\|\vec{P}_k - \vec{P}_a\|} \tag{14}$$

This placement satisfies the rigid-prosthesis constraint at every instant. The parameter d is user-configurable and must remain below the minimum shank segment length over the gait cycle. For the configuration used in this work, $d = 36.5$ cm, comprising a 14.5 cm residual limb and a 22 cm prosthesis shaft. With $L_{\text{shank}} = 0.246\,H$, this configuration is valid only for $H > 148.4$ cm.

2.8 Stance-phase export

Only the stance phase of the mounting-point trajectory is exported because the physical simulator characterizes GRF output, which requires ground contact. The coordinates of $\vec{P}_m$ are written to a comma-separated position-command file at a fixed 0.01 s time step. Values are quantized to the actuator resolution, which is 0.0125 cm horizontally, 0.00625 cm vertically, and 0.009° for the joint angle.

2.9 Operator interface and control-layer integration

A single-board computer running an operator interface was added to the simulator's embedded motor-control architecture, completing the chain from the position-command file of Section 2.8 to the simulator's actuators. The interface provides five runtime capabilities. It uploads and executes a newly generated position-command file, and it switches between simulated statures across successive trials without manual reconfiguration of the control stage. It also sets the force platform's mechanical tare reference, the vertical offset for a given trial, and the playback speed scale. A playback speed of 1/20 of normal walking speed was selected to avoid damage to the mechanical system. All 400 trials of this study were completed at this scale without mechanical damage under repeated cycling. Characterizing other scales is left for future work.

2.10 Evaluation protocol

2.10.1 Validation datasets

All reference datasets contain recordings from participants without lower-limb amputation, because the simulator reproduces able-bodied reference kinematics as a loading standard. The Kuopio dataset comprised 51 subjects; four (IDs 07, 09, 10, and 24) were excluded because their motion-capture files lacked the marker required to compute the shank-angle reference, yielding $N = 47$ (stature 161–187 cm). Hip, knee, and ankle positions were the joint centers distributed with the Kuopio dataset. Thigh and shank angles were computed from these joint centers in the sagittal plane with the four-quadrant arctangent. Frames flagged as occluded were discarded, and no additional filtering was applied. Initial contact was detected kinematically as the maximum anterior position of the ankle relative to the hip. This adapts the coordinate method of Zeni, Richards, and Higginson (2008), which uses the heel and sacrum markers,

to the joint centers used here. Each cycle was resampled to 101 points. The Senden et al. (2024) and Ferber et al. (2024) datasets were used exclusively for candidate model selection (Section 2.5.1) and do not appear in Table 5. The Ferber et al. (2024) dataset was used because it provides raw three-dimensional marker data from rigid clusters on the thigh and shank, together with stature and body mass measured for each participant. Its walking metadata list 2088 recordings from 1686 participants. Recordings were kept when stature, body mass, and sex were reported, and the injury severity field indicated "no injury". Recordings with a calculated walking speed outside 0.8 to 1.8 m/s were removed. The first recording listed for each participant was retained. Participants with a stature outside 140 to 210 cm were removed as out-of-range entries. The same was done for a body mass outside 30 to 150 kg. This left 456 participants. Within each sex, participants were ranked by stature, and 20 were taken at evenly spaced ranks. Stature in this sample ranged from 150 to 195 cm. Body mass ranged from 47.7 to 104.5 kg. Thirty-four of the 40 met the dataset's own definition of an uninjured participant. The other six had "no injury" recorded in the injury severity field but had a specific injury diagnosis involving the knee, ankle, or hip. Knee and hip positions were reconstructed from the raw marker clusters by combining the segment rotations and local joint-center offsets computed by the processing code published by the dataset authors. Knee position was then expressed relative to the hip. The Kuopio dataset was used not only for calibration and internal validation but also to select models for ankle position and shank angle, as described in Section 2.5.1.

Table 5: Datasets used for evaluation.

Dataset	*N*	Walking surface	Role
Kuopio (Lavikainen et al., 2024)	47	Overground	LOSO calibration + internal validation
Camargo (Camargo, Ramanathan, Flanagan, & Young, 2021)	22	Overground	External validation (held out entirely)

2.10.2 External validation

External validation uses the Camargo et al. (2021) dataset at three levels of increasing complexity. Because the Froude-estimated speed of the 22 subjects, 5.0–5.5 km/h, exceeds the validated range of the Koopman cycle-duration regression (0.5–5.0 km/h), this comparison also tests the extrapolation declared in Section 2.4.

Angle-only comparison evaluates angle prediction in isolation. The hip, knee, ankle, and shank angles produced by the calibrated Koopman model (Section 2.5.2) are compared against reference angles from the Camargo dataset. The hip, knee, and ankle angles are the inverse-kinematics angles distributed with the dataset. The shank angle is computed from the lateral knee and ankle markers. Gait cycles are delimited by

the heel-strike events distributed with the dataset. Each subject's stature is estimated from the shank length measured in the static trial, divided by the shank fraction of Section 2.3.

Geometry-only comparison evaluates the double-pendulum reduction independently of the angle model, and it is run on the Kuopio dataset. Thigh and shank angles are taken from the rigid-body orientation of the thigh and shank marker clusters, so they do not depend on the joint centers. The anatomical axis of each segment is defined during the static calibration trial, from the functional hip center estimated using the SCoRE method to the knee center, and from the knee center to the ankle center. Segment lengths are the center-to-center distances measured in that static trial, because the tape-measured lengths in the dataset metadata are 2 to 4 cm shorter than these distances. The pendulum is anchored at the measured hip center, and the resulting virtual knee and ankle positions are compared against the measured knee and ankle centers. Frames with an occluded joint center or fewer than three visible markers on a cluster are identified as invalid. Resulting gaps of up to 100 ms are interpolated in time, whereas cycles containing longer gaps are discarded. In that dataset, the reduction is evaluated by comparing the resulting joint angles with those obtained from inverse kinematics. Any observed bias is reported as a difference in conventions between inverse kinematics and marker-based geometry.

Full-pipeline comparison uses stature as its sole input and evaluates the predicted mounting-point trajectory against a reference trajectory derived from the Camargo marker data.

2.11 Prosthetic limb assembly

The residual limb is a three-layer surrogate built for this simulator. The upper layer, 8.5 cm of PETG HF (glycol-modified polyethylene terephthalate, high-flow grade), forms the main structural body and was selected over PLA (polylactic acid), which showed fractures under repeated compressive loading in earlier tests. Its upper face, 160 cm^2 in area, couples to the shank inclination axis of the simulator. The middle layer, 5 cm of Shore A10 silicone, is the only deformable element in the assembly and is intended to approximate the mild compression of soft tissue during the stance phase of human gait. The lower layer, 1 cm of TPU (thermoplastic polyurethane), prevents direct rigid contact between the silicone and the pyramid adapter and deforms slightly to help spread the load at that interface. Infill densities of 70 % for the PETG HF layer and 65 % for the TPU layer were selected empirically. The lower face of the assembly, 50 cm^2 in area, receives the pyramid adapter, which connects through a 22 cm shaft to a passive articulated prosthetic foot. The three-layer structure and the full assembly are shown in Fig. 2.

2.12 GRF characterization protocol

GRF signals were acquired using an AMTI OPT400600-2000 force platform connected to an AMTI Optima amplifier. The factory calibration of the force platform (serial number 9544) was verified using AMTI GEN5 software during these sessions. Data were recorded on a laptop running AMTI NetForce and exported to plain-text format

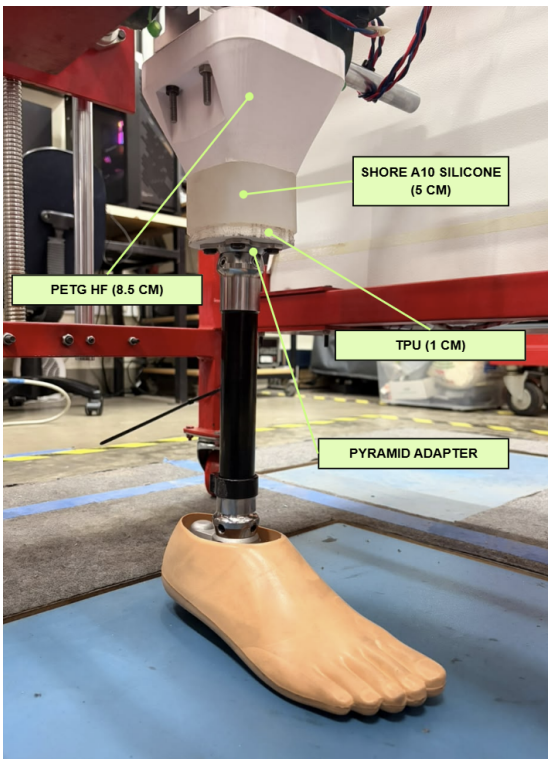


Fig. 2: Three-layer residual limb used in the simulator, with its pyramid adapter connected through a 22 cm shaft to the articulated prosthetic foot

using a custom Python script. Stance detection and the per-trial impulse-equivalent mass were computed in MATLAB R2025b on the same laptop. The regression models and the reliability statistics were computed in Python 3.14.6 with the NumPy and SciPy libraries. The same laptop also generates the position-command file. Fig. 3 shows the complete experimental setup.

Ten simulated statures were evaluated at 162, 164, 166, 168, 170, 172, 174, 176, 178, and 180 cm. For each stature, the stance-phase position-command file generated by the pipeline was transferred to the single-board computer and selected for execution. A reference position was established 1 mm above the platform surface, matching the tare position with no contact load. Five vertical offsets δ were then applied below this reference at 1, 2, 3, 4, and 5 mm. The smallest offset of 1 mm brought the foot to the platform surface, the nominal threshold of initial contact. Offsets of 2–5 mm corresponded to 1–4 mm of further nominal compression at this reference pose. During stance, the commanded trajectory lowers the mounting point below this reference by several additional millimeters as the limb advances. Therefore, the load at a given offset reflects the combined compression resulting from the commanded descent and the static offset, rather than from the offset alone. Each stature-offset combination was executed five consecutive times, for a total of 250 trials.

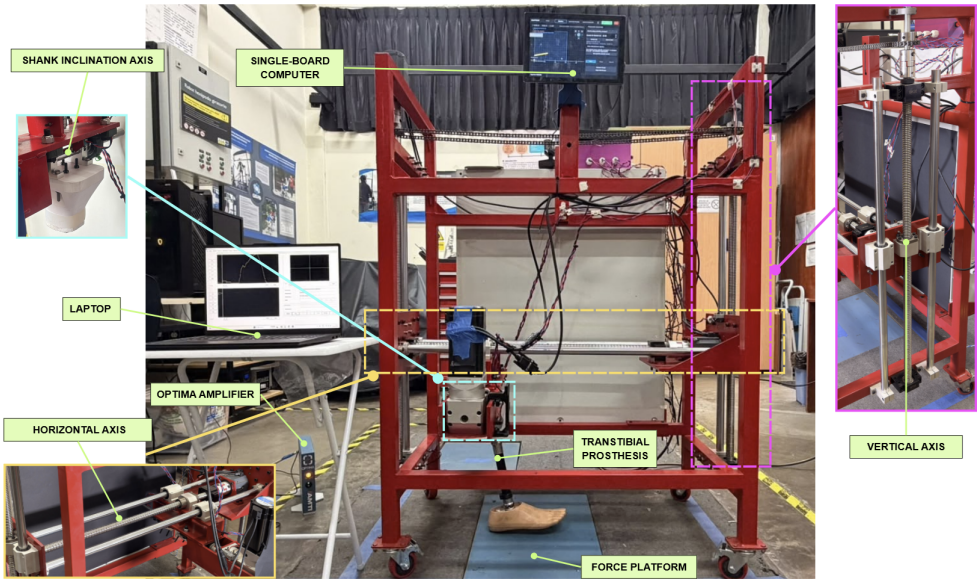


Fig. 3: Experimental setup for GRF characterization, showing the force platform, amplifier, gait simulator with mounted prosthetic assembly, control laptop, and single-board computer

The 250 trials were distributed across two sessions with a full simulator restart between sessions. Day 1 covered simulated statures from 162 to 170 cm (125 trials), and Day 2 covered 172 to 180 cm (125 trials). A third session reproduced statures 162, 170, and 180 cm with the complete five-offset protocol, adding 75 trials for inter-session reproducibility, for a total of 325 trials.

A prescriptive check used the inverse model, Eq. (17), to determine the required vertical offset for a fixed target body mass across fifteen simulated statures ranging from 162 and 180 cm, with five trials at each stature. The target mass was 60 kg. It was chosen so that the required offset fell within the calibrated range of 1 to 5 mm at every stature. This adds 75 trials.

The vertical ground reaction force (F_z) from each trial was recorded at 1000 Hz; no additional filtering was applied beyond the platform's internal anti-aliasing stage. Before each stature block, the platform was zeroed at the 1-mm reference position, which served as the tare for that block. Stance-phase onset and offset were detected by a 20 N threshold on the F_z signal (Zahradka et al., 2020). Trials in which the threshold was not crossed or in which contact was interrupted, detected as a gap of more than 10 consecutive samples below threshold within the expected stance window, were excluded and repeated. T_{stance} was measured as the real elapsed time between onset and offset detections. Each stance-phase record was then time-normalized to 100 % of its duration by interpolation to 101 equally spaced points (Kadaba, Ramakrishnan, & Wootten, 1990).

The impulse-equivalent mass m_{imp} is defined as the body mass of a person whose stance-phase support would deliver the same vertical impulse per gait cycle. In symmetric bipedal gait at steady state, both limbs together support the full body weight; the vertical impulse accumulated during a single stance phase therefore equals half the product of body weight and full gait-cycle duration. Solving for mass gives

$$m_{\mathrm{imp}} = \frac{2 \int_{\mathrm{stance}} F_z \, \mathrm{d}t}{g \, T_{\mathrm{cycle}}}, \tag{15}$$

where $g = 9.81\,\mathrm{m\,s^{-2}}$ and T_{cycle} is estimated as $T_{\mathrm{stance}}/0.60$, the normative stance fraction for level walking at comfortable speed (Perry & Burnfield, 2010). Substituting this estimate of T_{cycle} into Eq. (15) reduces m_{imp} to $1.2\,\overline{F_z}/g$, a scaled mean stance-phase force. The simulator is a position-controlled device that carries no body weight and moves quasi-statically, so m_{imp} does not represent a mass carried by the device. It expresses the mean stance force as a body mass, which is the quantity needed to prescribe a target load. Thus, m_{imp} is an operational loading metric rather than a measure of biomechanical equivalence to human gait. This formulation requires no normative peak-force coefficients and is independent of waveform shape.

Intra-session repeatability was quantified by the coefficient of variation (CV = $\mathrm{SD}/\bar{m}_{\mathrm{imp}} \times 100\,\%$) of the five m_{imp} values per stature-offset combination. Inter-session reproducibility was assessed using the intraclass correlation coefficient ICC(2,1), and within-condition reliability of the five-repetition mean was assessed using ICC(2,k). Both followed the thresholds of Koo and Li (2016), with values above 0.90 regarded as excellent. They were computed on the m_{imp} values of the 15 stature-offset combinations reproduced on Day 3, which comprised three statures by five offsets. A single measure per session was defined as the mean of the five repetitions. Day 3 data were not included in the regression. Inter-session agreement was further examined using Bland-Altman analysis (Bland & Altman, 1986), in which the differences between Day 3 and session-mean values were plotted against their corresponding means. Four candidate ordinary least-squares models were compared on the 45 session-mean values from Days 1 and 2. These values covered nine statures by five offsets. The 166 cm block was excluded because its m_{imp} showed a larger upward shift between the 1 mm and 2 mm offsets than at any other stature, and this shift persisted up to the 5 mm offset, as detailed in Section 3.3. The four models were offset only, stature only, additive combining offset and stature, and additive with an offset-by-stature interaction term. Model selection was based on the Akaike information criterion (AIC) (Burnham & Anderson, 2002), with lower values indicating a more favorable balance between model fit and complexity. For AIC comparison, the root-mean-square error (RMSE) was computed as $\sqrt{\mathrm{SS}_{\mathrm{res}}/n}$, where $n = 45$ is the number of observations. This formulation is consistent with the maximum-likelihood estimate of residual variance used in the AIC calculation.

The simulator executed the stance-phase trajectory at 1/20 of normal human walking speed, resulting in quasi-static platform loading. This assumes rate-independent contact stiffness between the prosthetic foot and the platform.

Each trial includes a pre-cycle period caused by the simulator's control architecture. While the high-level controller transfers the gait trajectory to the execution module, the simulator remains at the prescribed offset position, producing a sustained F_z level referred to as the pre-cycle plateau. Once the transfer is complete, the simulator moves 1 cm upward and 1 cm backward to break contact with the platform, then returns to the trajectory start position and executes the stance trajectory continuously, so that stance starts from no contact as it does in walking. This break-contact maneuver guarantees a load-free interval between the plateau and the gait cycle in every trial.

In the prescriptive session, the pre-cycle plateau of each trial was compared with the Day-1/2 calibration plateau curve. Linear interpolation was used to estimate the actual offset delivered, independently of the force-impulse mass estimate.

3 Results

3.1 Trajectory generator performance

Because the Kuopio Gait Dataset (Lavikainen et al., 2024) was also used for model selection and calibration, the LOSO results below quantify fit to the calibration population. Table 6 compares the generator with the population-mean baseline, and Figure 4 shows the mean trajectory envelopes.

Table 6: Leave-one-subject-out cross-validation against the Kuopio Gait Dataset ($N = 47$). The population-mean baseline predicts each left-out subject with the mean of the remaining 46 subjects, without any stature adjustment. RMSEnorm is defined in Section 2.2. RMSE is in cm for positions and in degrees for the shank angle. Values are mean ± SD across subjects.

Variable	r	RMSE	RMSEnorm	Class.	Baseline RMSEnorm
Knee X	0.999 ± 0.001	11.7 ± 6.6	1.31 ± 0.35	Good	0.89 ± 0.48
Knee Y	0.913 ± 0.042	1.0 ± 0.4	1.22 ± 0.48	Good	0.87 ± 0.52
Ankle X	0.999 ± 0.001	10.0 ± 5.5	1.39 ± 0.32	Good	0.91 ± 0.45
Ankle Y	0.985 ± 0.012	1.8 ± 0.7	1.14 ± 0.52	Good	0.90 ± 0.46
Shank angle	0.990 ± 0.009	5.3 ± 2.1	0.97 ± 0.36	Excellent	0.93 ± 0.40

Horizontal knee and ankle position achieved $r = 0.999$ across all 47 subjects, matching the r of the population-mean baseline. Temporal warping followed by PAVA produced forward motion with no backward steps in every stride. The shank angle yielded the lowest RMSEnorm (0.97), close to the baseline value of 0.93. Vertical knee and ankle displacement reached RMSEnorm between 1.14 and 1.22, above the corresponding baseline values of 0.87–0.90. The stature-scaled baseline gave RMSEnorm of 0.87 to 0.91 for the four positions, within 0.01 of the unscaled baseline. Stature scaling does not affect joint angles; therefore, the shank-angle RMSEnorm remains at 0.93. The generator's RMSEnorm was higher than that of both baselines in all five variables. In centimeters, the horizontal RMSE of the generator and of the scaled baseline differed by less than 0.2 cm. The rigid shank segment primarily affects the vertical

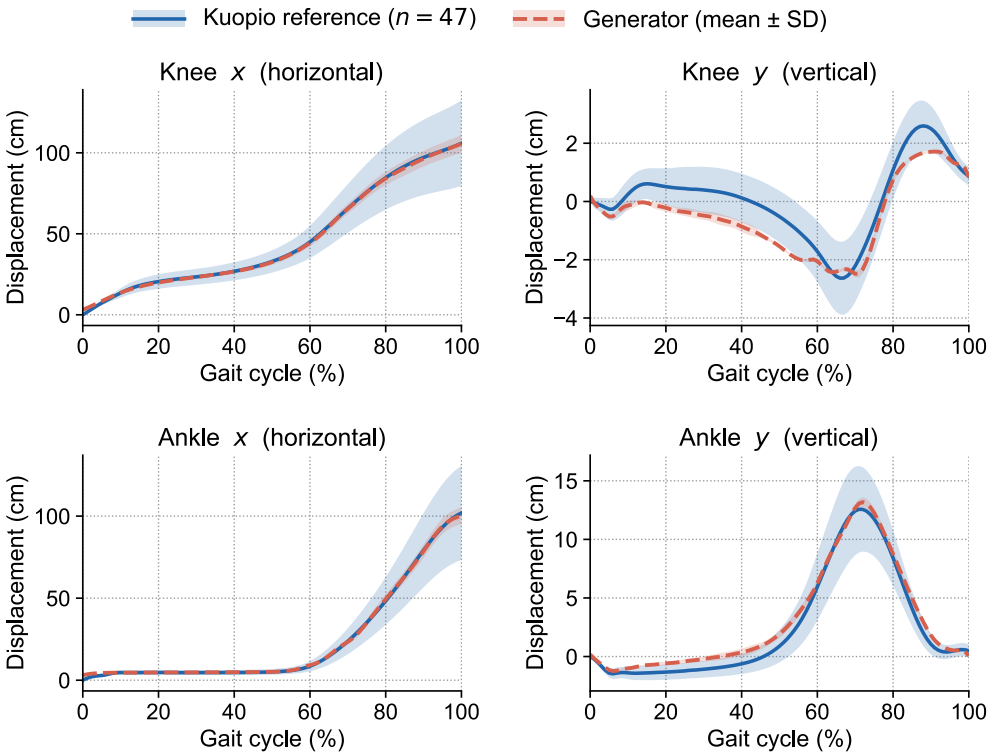


Fig. 4: Leave-one-subject-out cross-validation against the Kuopio Gait Dataset ($N = 47$). Shaded bands show mean ± SD across subjects. Blue: reference trajectories extracted from motion-capture data. Red dashed: generator output using stature as the sole input

components of the results obtained with the Kuopio dataset. Without the rigidity constraint, knee Y reached $r = 0.960$. Its RMSEnorm was 0.83. Ankle Y reached $r = 0.989$. Its RMSEnorm was 0.88. The horizontal correlations remain unchanged to three decimal places, and the horizontal RMSEnorm differs by no more than 0.003.

3.2 External validation

The Camargo dataset (Camargo et al., 2021) was held out from every calibration and model-selection step. Table 7 reports the angle-only comparison and Table 8 the full-pipeline comparison. At the joint-angle level, hip flexion and shank angle reached the Acceptable class, and the ankle angle showed the lowest agreement, with $r = 0.579 \pm 0.182$. The Camargo inverse-kinematics model defines knee flexion as negative, so the reduction of Eq. (3) becomes $\theta_{\text{shank}} = \varphi_{\text{hip}} + \varphi_{\text{knee}}$ with those angles. It yielded $r = 0.997 \pm 0.002$ when compared with marker-derived shank angles. RMSEnorm = 2.64 reflects a systematic bias between the inverse-kinematics and marker-geometry angle conventions and not a shape error. A shank of measured length placed at these angles

Table 7: External validation against the Camargo dataset, Angle-only comparison (joint angles, $N = 22$). RMSE is in degrees. Values are mean ± SD across subjects.

Joint	r	RMSE	RMSEnorm	Class.
Hip flexion	0.915 ± 0.033	13.8 ± 5.6	1.88 ± 0.74	Acceptable
Knee angle	0.834 ± 0.084	11.4 ± 2.5	2.32 ± 0.49	Deficient
Ankle angle	0.579 ± 0.182	11.7 ± 1.9	3.89 ± 0.65	Deficient
Shank angle	0.916 ± 0.053	7.4 ± 2.3	1.87 ± 0.54	Acceptable

Table 8: External validation against the Camargo dataset. Full-pipeline comparison with shank rigidity enforced ($N = 22$). RMSE is in cm. Values are mean ± SD across subjects.

Variable	r	RMSE	RMSEnorm	Class.
Knee X	0.996 ± 0.001	12.1 ± 4.9	1.82 ± 0.44	Acceptable
Knee Y	0.718 ± 0.102	1.4 ± 0.3	2.59 ± 0.54	Deficient
Ankle X	0.993 ± 0.005	17.6 ± 5.7	2.38 ± 0.76	Deficient
Ankle Y	0.908 ± 0.061	2.1 ± 0.5	2.86 ± 0.56	Deficient

from the knee marker gave a mean ankle bias of -41.7 mm in X and $+16.1$ mm in Y over 22 participants. This bias is consistent with a difference in angle conventions, as the same geometric construction using marker-derived angles yielded an RMSE of 2.4 mm in X and 3.4 mm in Y with biases below 1 mm.

In the geometry-only comparison on Kuopio, which used 391 gait cycles from 47 participants, the pendulum reproduced the measured knee center with an RMSE of 9.4 ± 11.0 mm in X and 6.3 ± 4.5 mm in Y, and the measured ankle center with an RMSE of 13.4 ± 12.0 mm in X and 10.9 ± 5.9 mm in Y. Mean biases were -1.6 mm and $+1.9$ mm at the knee and -0.6 mm and -5.7 mm at the ankle, in X and Y respectively. With the normalization used in Table 6, RMSEnorm was 0.21 and 0.74 for the knee and 0.99 and 1.11 for the ankle. The largest errors came from one participant whose thigh-cluster orientation differed by 7° from the static joint-center axis, which points to cluster displacement after the static trial rather than to the pendulum geometry.

In the full-pipeline comparison, position correlation was $r \geq 0.993$ in X and 0.718 to 0.908 in Y, where both joints fell in the Deficient class. In the model, joint displacement scales exactly with stature. In the Camargo data, displacement correlated only weakly with stature, with $r = 0.33$–0.35 and $p > 0.1$. The mean shank-angle bias in Camargo decreased from $+8.7°$ before Kuopio calibration to $-1.1°$ after calibration. The between-subject standard deviation of the shank angle was 3.3° in Camargo. The corresponding value in Kuopio was 5.2°. The Kuopio mean, scaled by stature for the positions, was also applied to the 22 Camargo subjects. Its RMSEnorm was 2.00 to 3.70 across the same five variables. The generator had lower RMSEnorm in four variables and higher RMSEnorm for vertical knee position. In centimeters, the mean had lower RMSE for both horizontal positions and for vertical knee position. Neither source was consistently more accurate on the external data.

3.3 GRF characterization

Within-block coefficient of variation of m_{imp} over the five repetitions had a median of 0.17 % (≈0.1 kg) and a maximum of 0.80 % (≈0.5 kg) across the 45 stature–offset combinations of Days 1 and 2 used in model fitting. Figure 5 shows the mean force-time profiles of the 50 stature-offset combinations, each averaged over five repetitions.

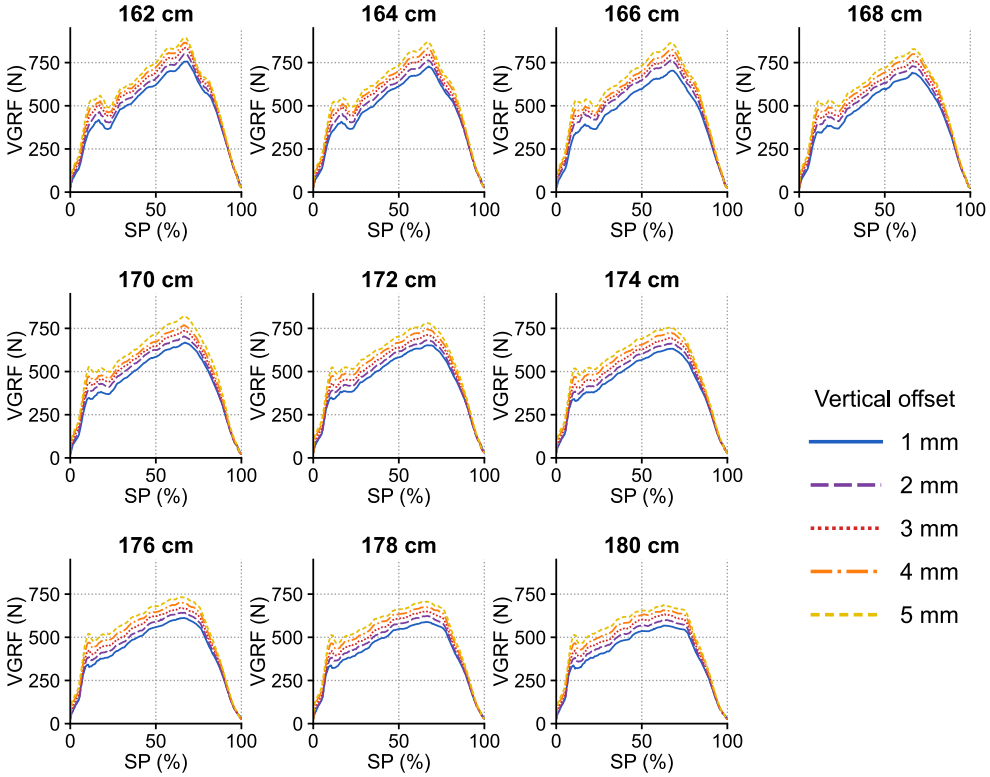


Fig. 5: Mean vertical ground reaction force (VGRF) profiles across the stance phase (SP) for all ten simulated statures (162–180 cm). Each panel shows mean curves over five repetitions for five vertical-offset conditions, identified by color and line style: blue solid (1 mm), purple dashed (2 mm), red dotted (3 mm), orange dash-dotted (4 mm), and yellow short-dashed (5 mm). VGRF is expressed in newtons (N), and SP is expressed as a percentage (%)

In the initial fit using all 50 combinations, the 166 cm block exhibited a level shift relative to the other statures, emerging between the 1 mm and 2 mm offsets and persisting through the 5 mm offset. Its step between these two offsets was 5.8 kg, compared with 3.2 to 4.2 kg at the other statures. It also produced the largest single residual of +2.67 SD at $\delta = 2$ mm, although the five repetitions ranged from only 59.9 to 60.4 kg. The block was recorded without interruption, and its pre-cycle plateau and

stance duration were in line with the neighboring statures. Between the last 1-mm run and the first 2-mm run, the mediolateral center of pressure during stance moved by 5.2 mm. This step was more than 30 times the spread between repetitions and coincided with the level shift, suggesting a change in the contact conditions between the prosthetic foot and the platform. The residual did not qualify as an outlier after Bonferroni correction. The exclusion was therefore based on the observed discontinuity and coincident center-of-pressure shift, rather than on a statistical outlier criterion. The block was excluded from the final regression, leaving 45 combinations, and is shown as a dashed line in Fig. 6. When the additive model was fitted to all 50 combinations, the stature and offset slopes changed by less than 3 %, the additive model retained the lowest AIC among the four candidates, and the prescriptive-session bias changed by about 0.1 kg. These results show that including the excluded block had a limited effect on the fitted coefficients, model selection, and prescriptive-session bias. When fitted to the 45 remaining combinations, the additive model again yielded the lowest AIC among the four candidate models. It is given by Eq. (16),

$$m_{\mathrm{imp}} = 135.2230 - 0.5019H + 3.2707\delta, \tag{16}$$

with H in cm and δ in mm, valid for $H \in [162, 180]$ cm (excluding 166 cm) and $\delta \in [1, 5]$ mm, and it reached $R^2 = 0.993$ with a residual standard error of 0.49 kg, as shown in Fig. 6. The 95 % confidence intervals were 130.9 to 139.6 kg for the intercept, -0.527 to -0.477 kg/cm for stature, and 3.17 to 3.37 kg/mm for offset. These conventional ordinary least-squares confidence intervals should be interpreted cautiously because of the residual dependence described below. A Shapiro-Wilk test did not reject normality of the residuals, with $W = 0.965$ and $p = 0.18$. The residuals were, however, autocorrelated across the ordered grid and grouped by stature block, so this p value is approximate. The Durbin-Watson statistic was 0.76. This indicates structure that the linear form does not capture. Adding the interaction term yielded $\Delta\mathrm{AIC} = 1.6$ and did not reduce the residual error, providing no evidence of an improvement in model fit from allowing the offset slope to vary with stature. The offset-only and stature-only models had ΔAIC values of 163 and 204, respectively. Table 9 lists the four candidate models.

Table 9: Candidate ordinary least-squares models for m_{imp} on the 45 session-mean values of Days 1 and 2, with 166 cm excluded. k is the number of fitted coefficients, including the intercept. RSE denotes the residual standard error, calculated using $n-k$ residual degrees of freedom. ΔAIC is relative to the additive model.

Model	k	R^2	RSE (kg)	ΔAIC
Offset only	2	0.711	3.01	163.3
Stature only	2	0.281	4.76	204.4
Additive	3	0.993	0.49	0.0
Additive with interaction	4	0.993	0.49	1.6

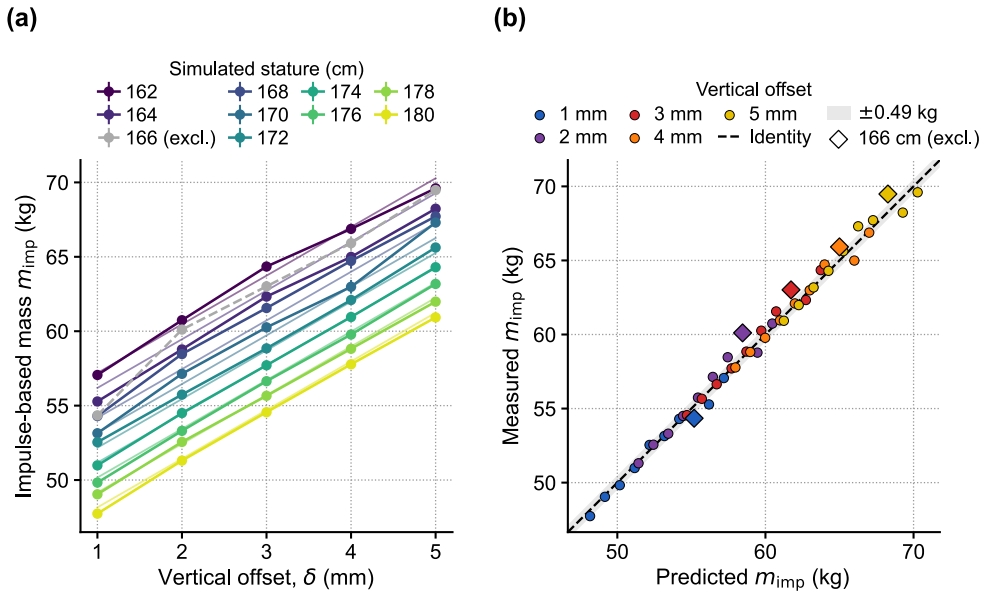


Fig. 6: Impulse-equivalent mass model. (a) m_{imp} as a function of vertical offset δ for ten simulated statures; points show mean ± SD over five repetitions, thick solid lines connect the measured points, thin solid lines show the additive model fit, and the excluded 166 cm block is shown gray and dashed. (b) Predicted vs. measured m_{imp}, colored by offset (166 cm block as outlined diamonds), with the ±0.49-kg residual standard error shown as a gray band; the dashed line is the identity

At fixed offset, each additional centimeter of simulated stature corresponds to a 0.50 kg decrease in m_{imp}. Across the 162–180 cm range, this corresponds to a total decrease of approximately 9.0 kg. By inverting the fitted model, the vertical offset required to achieve a target impulse-equivalent mass m_{target} at stature H is

$$\delta = \frac{m_{\mathrm{target}} - 135.2230 + 0.5019\,H}{3.2707}. \tag{17}$$

The coefficients are reported to four decimal places because they were used in Eq. (17) during the prescriptive session. The computed offset was rounded to 0.1 mm, the resolution of the vertical-offset setting.

Across the 15 Day-3 conditions, the mean difference between sessions was +0.080 kg, with 95 % limits of agreement from −0.988 to +1.148 kg (Bland & Altman, 1986), equivalent to about ±0.3 mm of vertical offset (Fig. 7). Within-condition reliability of m_{imp} gave ICC(2,k) > 0.999, with the lower bound of its 95 % confidence interval also above 0.999. Inter-session reproducibility, comparing Day 3 against the corresponding Day-1/2 measurements, gave ICC(2,1) = 0.996, with a 95 % confidence interval of 0.989 to 0.999. Both lower bounds exceed the 0.90 threshold for excellent

reliability (Koo & Li, 2016). However, both coefficients are computed across conditions spanning roughly 48 to 70 kg. With such a large between-condition variance, a high ICC is expected. Therefore, the absolute limits of agreement reported above provide a more informative measure of inter-session reproducibility. At 170 cm, recorded on Day 1, all five Day-3 differences were positive. At 180 cm, recorded on Day 2, all five were negative, between −0.20 and −0.60 kg. Within Day 3 alone, the stature slope was −0.48 kg/cm, linear across both halves of the range.

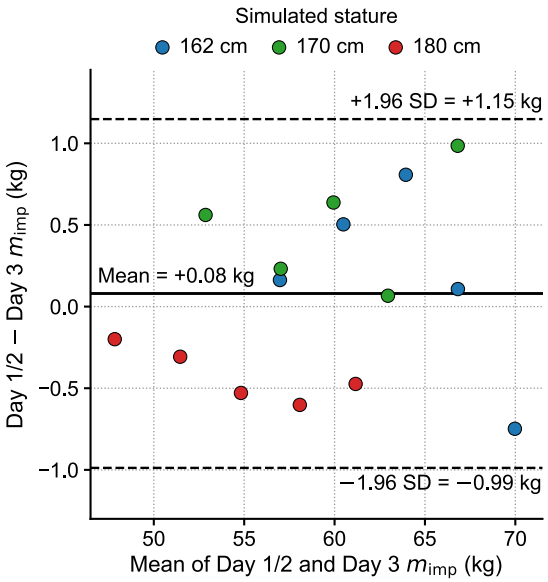


Fig. 7: Bland-Altman plot of inter-session agreement for m_{imp}, comparing Day 1/2 session means against Day 3

Of the fifteen prescriptive statures, nine were on the original 2 cm calibration grid spanning 162–180 cm, excluding 166 cm. One was the 166 cm stature excluded from the regression fit. The other five were additional statures, 163, 165, 167, 173, and 177 cm, obtained by interpolation of the fitted model and never used in fitting it. Figure 8 shows the signed deviation $m_{\text{imp}} - m_{\text{pred}}$ at each stature, labeled by grid membership. Across the eight calibration-grid statures 164–180 cm, the mean m_{imp} exceeded the 60-kg target by 2.1 kg. Pre-cycle plateau analysis attributed part of this deviation to a between-session tare shift of about +0.36 mm. Through the model slope of 3.27 kg/mm, that shift accounts for about 1.2 kg of the 2.1-kg deviation. Four of the five interpolated statures showed a comparable bias of 2.0 to 2.1 kg. The fifth, 177 cm, showed a higher bias of 3.3 kg. The excluded 166 cm stature showed a bias of 2.3 kg, within the range of the calibration grid. If the departure seen in calibration had been specific to that stature, its bias would have been about 3.5 kg, so the anomaly did not recur when the stature was measured again. Intra-condition repeatability across all fifteen statures had a median SD of 0.06 kg and a maximum of 0.16 kg, unchanged from

the nine-stature grid alone. At 162 cm, m_{imp} fell 1.6 kg below target, attributable to a tare shift of −0.76 mm in the opposite direction, confirmed by the pre-cycle plateau.

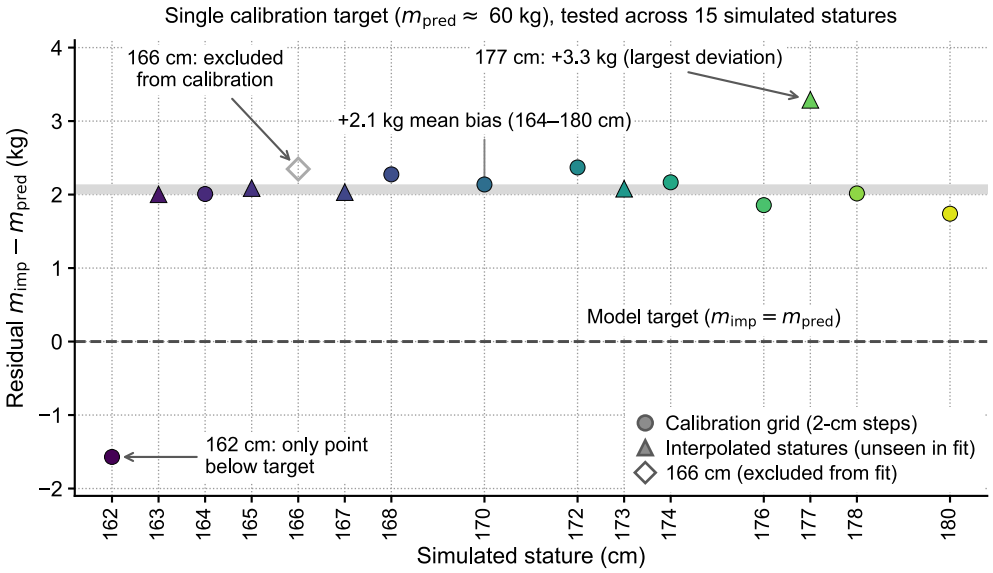


Fig. 8: Residual mass error ($m_{\text{imp}} - m_{\text{pred}}$) across all fifteen tested statures in the prescriptive session, with a single target mass ($m_{\text{pred}} \approx 60$ kg) used at each stature. The dashed line marks the model target; the shaded band marks the mean bias across the calibration-grid statures. Marker color encodes stature; marker shape distinguishes the calibration grid (circles), interpolated statures (triangles), and 166 cm (open diamond)

4 Discussion

4.1 Trajectory generator

For the Kuopio dataset, the generator's error was higher than that of the population-mean baseline across all five variables, with or without stature scaling of the mean. Stature probably explains little of the kinematic variability within a cohort, and a stature-driven model cannot do much better than a mean reference for subjects from that cohort. For each target stature, the generator provides a forward-moving trajectory that preserves shank rigidity and can be executed by the simulator, achieving a Good or better accuracy classification on the Kuopio dataset. A stature-scaled mean trajectory with the same rigidity constraint could serve the GRF protocol equally well.

Speed extrapolation does not explain the angle degradation observed in the Camargo dataset, because the angle model uses a fixed reference speed and stature, yielding the same raw angles for every subject. The Kuopio calibration removed most of the shared shank-angle bias in Camargo, and the remaining error varies between

subjects. The Camargo reference also varies less between subjects than the Kuopio reference, which raises RMSEnorm.

The generator cannot reproduce real inter-subject variability. With stature as its only input, its shank-angle variation between subjects is a few tenths of a degree. In Kuopio, the between-subject standard deviation of the shank angle averaged 5.2° over the cycle. Real shank angle showed weak correlations with stature or speed, suggesting that individual gait style may contribute to the observed variability. However, reproducing inter-subject variability is not required for its intended use of providing a reproducible reference condition across statures.

To our knowledge, none of the published trajectory generators, data-driven or analytical, has been integrated directly with a physical prosthetic gait simulator. The simulator prescribes motion rather than force; therefore, the resulting ground reaction force (GRF) must be characterized empirically.

4.2 GRF characterization

Because the interaction term did not improve model fit, the additive calibration model was retained for the evaluated stature range of 162–180 cm. The decrease in load with stature has a geometric cause, not a physiological one. Taller simulated statures raise the ankle trajectory earlier and further during mid-to-late stance, so the mounting point descends less below the reference pose and the rigid-length prosthesis is compressed less. A different simulator, foot, or mounting distance would need its own calibration run.

The 95 % limits of agreement correspond to about ±0.3 mm of vertical offset, representing the practical precision floor for prescriptive use. The mean inter-session difference of +0.080 kg is smaller than the half-width of the agreement interval, but its sign depends on the calibration session. We attribute this dependence to a small tare shift between the two calibration sessions, consistent with the shifts identified from the pre-cycle plateau. This session dependence does not account for the stature effect, because Day 3 measured 162, 170, and 180 cm within a single session and its stature slope was close to the calibration model's value of −0.50 kg/cm.

In the prescriptive session, the positive bias of 2.1 kg at the calibration-grid statures arose partly from the tare shift identified from the pre-cycle plateau. That shift accounts for about 1.2 kg of the deviation. The remaining 0.9 kg lies within the Days-1/2-to-Day-3 limits of agreement and cannot be distinguished from ordinary inter-session variability. The negative bias at 162 cm, also explained by a tare shift, suggests that tare drift is the largest identified source of prescriptive error. Within the prescriptive session, the median intra-condition standard deviation was 0.06 kg, so once the tare reference is established the simulator delivers stable repeated trials. The pre-cycle plateau provides a diagnostic of the tare state that does not depend on the impulse model, and it should be checked before each stature block in prescriptive use.

4.3 Limitations

The prescriptive GRF protocol characterizes the vertical force when the simulator executes a trajectory derived from able-bodied gait. The loading asymmetries of amputee

gait are not represented in the current reference trajectory and are the most clinically relevant extension for future work. The actual loading response depends on how the imposed trajectory interacts with the stiffness of the component under test, so the prescriptive characterization documents the reference condition but does not replace direct measurement on each specific component. Inter-session reproducibility is limited by manual repositioning of the tare reference. Among the five interpolated statures, 177 cm showed a residual of +3.3 kg, larger than in the rest of the tested range. Because this stature lacks a calibration plateau curve, the pre-cycle-plateau tare diagnostic used at the grid statures cannot be applied to it in the same way. A single condition cannot show whether this residual comes from the same tare drift or from another source of error. Repeating measurements at this stature across additional sessions would help distinguish between these possible sources of error. If the prosthetic foot is viscoelastic, the quasi-static execution used here may give a force magnitude different from that at physiological speed. This effect would act through foot contact, not through body inertia. As expected for this quasi-static, position-driven mechanism, the measured F_z waveform rises to a single peak at approximately 62–67 % of stance, instead of the two-peak profile with a mid-stance valley typical of human vertical GRF (Winter, 2009), and the normalized waveforms correlate with $r = 0.74$–0.79. Because Eq. (15) computes m_{imp} from the mean force over stance rather than the waveform shape, differences in waveform shape do not directly affect the calculated values. However, these differences indicate that the simulator does not fully reproduce the temporal loading pattern of human gait.

Only the vertical force was analyzed, although the platform also recorded the horizontal force components. A single prosthetic foot and one residual-limb configuration were tested. All tests used a fixed mounting-point distance of $d = 36.5$ cm, and the effect of varying this distance on the measured forces was not evaluated. The force model and the prescriptive check covered statures only between 162 and 180 cm. Within that range, stature is partly confounded with session date, because Days 1 and 2 each covered one half of it, but the single-session check on Day 3 limits this confound. The accuracy with which the simulator followed the commanded trajectory was not measured; therefore, the measured GRF cannot be directly associated with the prescribed trajectory without assuming accurate trajectory tracking.

The generator was selected, calibrated, and internally validated on the same Kuopio dataset, with LOSO testing to limit the overlap.

4.4 Future work

Public motion data from prosthesis users would enable the adaptation of the generator's calibration and position correction to amputee gait and the application of the GRF protocol under amputee loading. Testing other values of d would cover other residual-limb and prosthesis lengths.

A pending validation of the physical system will compare the commanded trajectory with the trajectory actually executed by the motors. The completed control-layer integration makes this comparison possible. With position and motor-torque sensing in a closed loop, the tracking error could be monitored and corrected during execution. Position feedback during trajectory execution could also improve inter-session

GRF reproducibility by reducing tracking errors, although variability associated with manual tare repositioning would remain. Embedding the trajectory generator and the inverse model Eq. (17) on the single-board computer would allow the operator interface to derive and apply the required vertical offset for a target body mass at runtime, without the current offline calculation step. A further extension is closed-loop force control, using the characterized offset-to-mass relationship or a dedicated force sensor to adjust the prescribed trajectory in real time. The simulator could then track a target GRF profile instead of a fixed position. Knowing the position-to-force relationship before a test matters most for components with a force or torque limit, such as active prostheses, where a load above the rated value could damage the actuator. The component tested in this work was a passive articulated foot, so this use of the protocol remains prospective.

5 Conclusion

This work presented a prescriptive GRF protocol for reproducible transtibial prosthesis evaluation, characterizing the vertical ground reaction force resulting from prosthetic foot–surface interaction. A stature-driven gait trajectory generator, built on published models and evaluated against public gait datasets, supplied the trajectories. A three-degree-of-freedom gait simulator positioned above a force platform executed these trajectories, enabling experimental calibration and evaluation of the GRF protocol. For the Kuopio dataset, the generator did not outperform the population-mean baselines; however, its primary role is to provide a simulator-executable trajectory for each target stature. The prescriptive protocol was repeatable within a session, with a median intra-condition standard deviation of 0.06 kg. In the prescriptive evaluation, its mean bias was +2.1 kg against the 60 kg target across eight calibration-grid statures (164–180 cm), equivalent to 3.5 % of the target. A between-session tare shift explained about 1.2 kg of it, and the remainder lay within the inter-session limits of agreement, from −0.988 to +1.148 kg. These limits define the usable precision of the prescriptive protocol, which also includes a session-to-session tare diagnostic procedure. Because the resulting GRF depends on the mechanical characteristics of the component under test, experimental characterization remains necessary for each prosthetic configuration. The resulting system enables repeatable, calibrated loading tests of prosthetic components without requiring amputee participants in every test session.

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Data availability. Data supporting the conclusions of this paper are available from the corresponding author upon request. Additional data and source code related to this work may also be requested from the corresponding author.

Ethics approval. This study collected no data from human participants. The gait datasets used for model selection, calibration and validation (Camargo et al., 2021; Ferber et al., 2024; Lavikainen et al., 2024; Senden et al., 2024) were previously published, each with ethics approval from the review board of its collecting institution.

Consent to participate. Informed consent was obtained from all participants by the original investigators of each dataset.

Declarations

Conflict of interest. The authors declare no competing interests.

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